

Kernel Engineering

From silicon to speculative decoding – GPU kernels for modern LLMs.

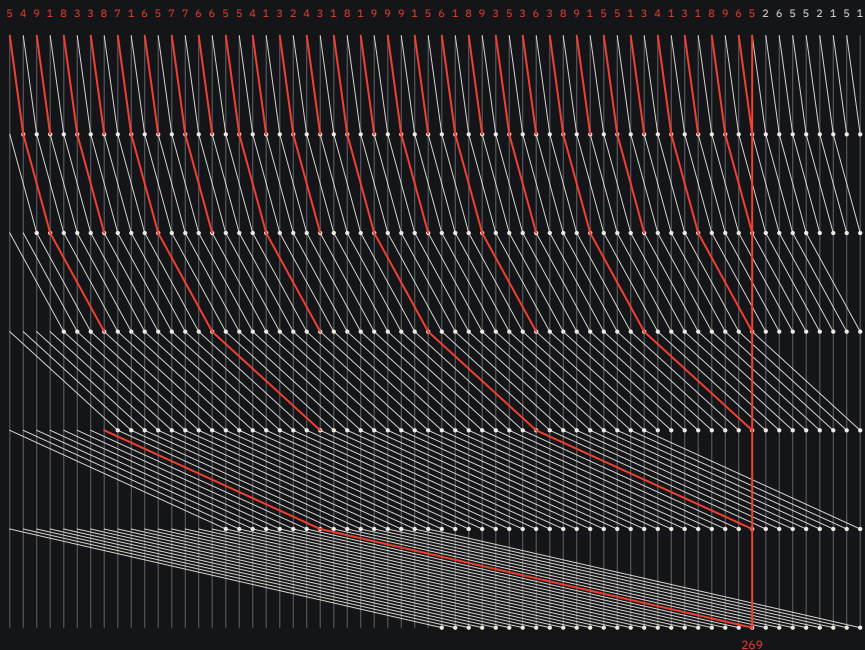


FIG. 0 A Kogge-Stone prefix sum, the pattern inside warp-level scans: 64 GPU lanes, 6 steps, 321 additions. At step k each lane adds the partial sum 2^k lanes to its left. In red: the additions lane 55 depends on, gathering 56 of the digits above into 269.